Tier 1 / Foundations

Read the wire; build a useful local server.

Mental model

Host = UI, model and permissions. Client = one protocol peer. Server = capabilities and allowed data.

Tools perform operations; resources expose content; prompts provide templates. The host chooses what enters model context.

First run

uv sync; npm ci

uv run python scripts/check_env.py

uv run python scripts/foundation_server.py (waits for protocol input)

npm run inspector -- uv run python scripts/foundation_server.py

Current exchange

Target 2026-07-28. Discover support with server/discover. Every request carries revision and client capabilities in reserved _meta keys.

Match response IDs to request IDs. Notifications have no ID and no response.

A complete tool result may still have isError=true.

Useful primitives

tools/list -> tools/call

resources/list or resources/templates/list -> resources/read

prompts/list -> prompts/get; completion/complete suggests an argument value.

Security habits

Keep stdout exclusively for JSON-RPC; log to stderr.

Validate argument shape and bounds at runtime. Tool annotations are hints, not authorization.

Require a separate review before a write; never let a model grant its own permission.

Checkpoint

uv run python scripts/tier1.py

Explain the request, response, resource URI and approval boundary.

Next: modules 07-10; details in docs/PROTOCOL_GUIDE.md.